## EXAMPLE PROBLEM 4.1

### Number-of-Vacancies Computation at a Specified Temperature

Calculate the equilibrium number of vacancies per cubic meter for copper at 1000°C. The energy for vacancy formation is 0.9 eV/atom; the atomic weight and density (at 1000°C) for copper are 63.5 g/mol and 8.4 g/cm$^3$, respectively.

***Solution***

This problem may be solved by using Equation 4.1; it is first necessary, however, to determine the value of $N_{Cu}$—the number of atomic sites per cubic meter for copper, from its atomic weight $A_{Cu}$, its density $\rho$, and Avogadro's number $N_A$, according to

Number of atoms per unit volume for a metal

$$N_{Cu} = \frac{N_A \rho}{A_{Cu}} \tag{4.2}$$

$$= \frac{(6.022 \times 10^{23}\ \text{atoms/mol})(8.4\ \text{g/cm}^3)(10^6\ \text{cm}^3/\text{m}^3)}{63.5\ \text{g/mol}}$$

$$= 8.0 \times 10^{28}\ \text{atoms/m}^3$$

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Thus, the number of vacancies at 1000°C (1273 K) is equal to

$$N_v = N \exp\left(-\frac{Q_v}{kT}\right)$$

$$= (8.0 \times 10^{28}\ \text{atoms/m}^3) \exp\left[-\frac{(0.9\ \text{eV})}{(8.62 \times 10^{-5}\ \text{eV/K})(1273\ \text{K})}\right]$$

$$= 2.2 \times 10^{25}\ \text{vacancies/m}^3$$

## 4.3 IMPURITIES IN SOLIDS

A pure metal consisting of only one type of atom just isn't possible; impurity or foreign atoms are always present, and some exist as crystalline point defects. In fact, even with relatively sophisticated techniques, it is difficult to refine metals to a purity in excess of 99.9999%. At this level, on the order of $10^{22}$ to $10^{23}$ impurity atoms are present in 1 m$^3$ of material. Most familiar metals are not highly pure; rather, they are **alloys,** in which impurity atoms have been added intentionally to impart specific characteristics to the material. Ordinarily, alloying is used in metals to improve mechanical strength and corrosion resistance. For example, sterling silver is a 92.5% silver–7.5% copper alloy. In normal ambient environments, pure silver is highly corrosion resistant, but also very soft. Alloying with copper significantly enhances the mechanical strength without reducing the corrosion resistance appreciably.

alloy

The addition of impurity atoms to a metal results in the formation of a **solid solution** and/or a new *second phase,* depending on the kinds of impurity, their concentrations, and the temperature of the alloy. The present discussion is concerned with the notion of a solid solution; treatment of the formation of a new phase is deferred to Chapter 9.

solid solution

Several terms relating to impurities and solid solutions deserve mention. With regard to alloys, **solute** and **solvent** are terms that are commonly employed. *Solvent* is the element or compound that is present in the greatest amount; on occasion, solvent atoms are also called *host atoms*. *Solute* is used to denote an element or compound present in a minor concentration.

solute, solvent

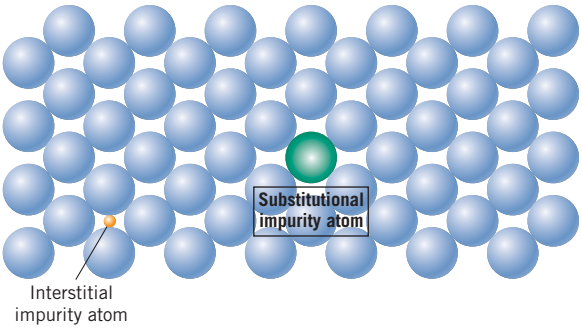



**Figure 4.2** Two-dimensional schematic representations of substitutional and interstitial impurity atoms. (Adapted from W. G. Moffatt, G. W. Pearsall, and J. Wulff, *The Structure and Properties of Materials*, Vol. I, *Structure*, John Wiley & Sons, 1964. Reproduced with permission of Janet M. Moffatt.)

## Solid Solutions

A solid solution forms when, as the solute atoms are added to the host material, the crystal structure is maintained and no new structures are formed. Perhaps it is useful to draw an analogy with a liquid solution. If two liquids that are soluble in each other (such as water and alcohol) are combined, a liquid solution is produced as the molecules intermix, and its composition is homogeneous throughout. A solid solution is also compositionally homogeneous; the impurity atoms are randomly and uniformly dispersed within the solid.

**substitutional solid solution**

**interstitial solid solution**

Impurity point defects are found in solid solutions, of which there are two types: **substitutional** and **interstitial.** For the substitutional type, solute or impurity atoms replace or substitute for the host atoms (Figure 4.2). Several features of the solute and solvent atoms determine the degree to which the former dissolves in the latter. These are expressed as four *Hume–Rothery rules,* as follows:

1. *Atomic size factor*. Appreciable quantities of a solute may be accommodated in this type of solid solution only when the difference in atomic radii between the two atom types is less than about ±15%. Otherwise, the solute atoms create substantial lattice distortions and a new phase forms.
2. *Crystal structure*. For appreciable solid solubility, the crystal structures for metals of both atom types must be the same.
3. *Electronegativity factor*. The more electropositive one element and the more electronegative the other, the greater the likelihood that they will form an intermetallic compound instead of a substitutional solid solution.
4. *Valences*. Other factors being equal, a metal has more of a tendency to dissolve another metal of higher valency than to dissolve one of a lower valency.

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**Tutorial Video: Defects**
What Are the Differences between Interstitial and Substitutional Solid Solutions?

An example of a substitutional solid solution is found for copper and nickel. These two elements are completely soluble in one another at all proportions. With regard to the aforementioned rules that govern degree of solubility, the atomic radii for copper and nickel are 0.128 and 0.125 nm, respectively; both have the FCC crystal structure; and their electronegativities are 1.9 and 1.8 (Figure 2.9). Finally, the most common valences are +1 for copper (although it sometimes can be +2) and +2 for nickel.

For interstitial solid solutions, impurity atoms fill the voids or interstices among the host atoms (see Figure 4.2). For both FCC and BCC crystal structures, there are two types of interstitial sites—*tetrahedral* and *octahedral;* these are distinguished by the number of nearest neighbor host atoms—that is, the coordination number. Tetrahedral sites

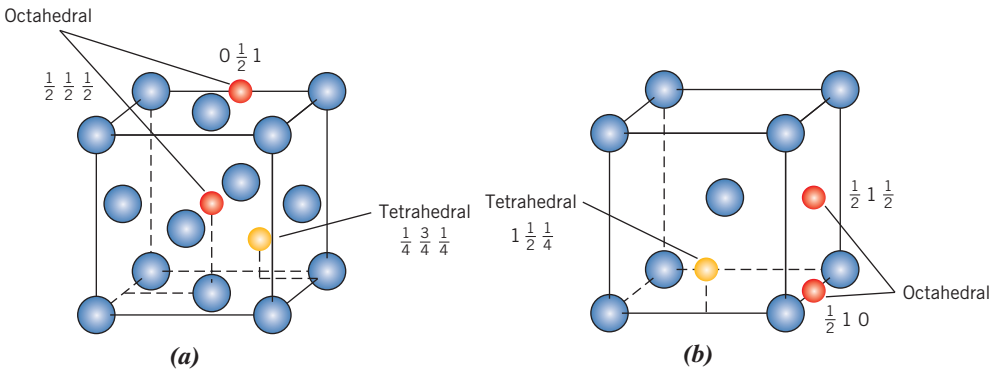



**Figure 4.3** Locations of tetrahedral and octahedral interstitial sites within (*a*) FCC and (*b*) BCC unit cells.

have a coordination number of 4; straight lines drawn from the centers of the surrounding host atoms form a four-sided tetrahedron. However, for octahedral sites the coordination number is 6; an octahedron is produced by joining these six sphere centers.[3] For FCC, there are two types of octahedral sites with representative point coordinates of $0\,\frac{1}{2}\,1$ and $\frac{1}{2}\,\frac{1}{2}\,\frac{1}{2}$. Representative coordinates for a single tetrahedral site type are $\frac{1}{4}\,\frac{3}{4}\,\frac{1}{4}$.[4] Locations of these sites within the FCC unit cell are noted in Figure 4.3*a*. Likewise, for BCC, there are two octahedral and one tetrahedral types. Representative coordinates for the octahedral sites are as follows: $\frac{1}{2}\,1\,\frac{1}{2}$ and $\frac{1}{2}\,1\,0$; for BCC tetrahedral, $1\,\frac{1}{2}\,\frac{1}{4}$ is a representative coordinate. Figure 4.3*b* shows the positions of these sites within a BCC unit cell.[4]

Metallic materials have relatively high atomic packing factors, which means that these interstitial positions are relatively small. Consequently, the atomic diameter of an interstitial impurity must be substantially smaller than that of the host atoms. Normally, the maximum allowable concentration of interstitial impurity atoms is low (less than 10%). Even very small impurity atoms are ordinarily larger than the interstitial sites, and as a consequence, they introduce some lattice strains on the adjacent host atoms. Problems 4.8 and 4.9 call for determination of the radii of impurity atoms $r$ (in terms of $R$, the host atom radius) that just fit into tetrahedral and octahedral interstitial positions of both BCC and FCC without introducing any lattice strains.

Carbon forms an interstitial solid solution when added to iron; the maximum concentration of carbon is about 2%. The atomic radius of the carbon atom is much less than that of iron: 0.071 nm versus 0.124 nm.

Solid solutions are also possible for ceramic materials, as discussed in Section 12.5.

## EXAMPLE PROBLEM 4.2

### Computation of Radius of BCC Interstitial Site

Compute the radius $r$ of an impurity atom that just fits into a BCC octahedral site in terms of the atomic radius $R$ of the host atom (without introducing lattice strains).

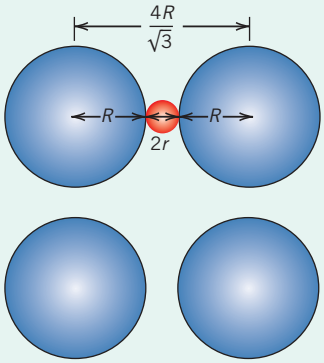
#### *Solution*

As Figure 4.3*b* notes, for BCC, one octahedral interstitial site is situated at the center of a unit cell edge. In order for an interstitial atom to be positioned in this site without introducing lattice

[3]The geometries of these site types may be observed in Figure 12.7.

[4]Other octahedral and tetrahedral interstices are located at positions within the unit cell that are equivalent to these representative ones.

strains, the atom just touches the two adjacent host atoms, which are corner atoms of the unit cell. The drawing shows atoms on the (100) face of a BCC unit cell; the large circles represent the host atoms—the small circle represents an interstitial atom that is positioned in an octahedral site on the cube edge.




On this drawing is noted the unit cell edge length—the distance between the centers of the corner atoms—which, from Equation 3.4, is equal to

$$\text{Unit cell edge length} = \frac{4R}{\sqrt{3}}$$

Also shown is that the unit cell edge length is equal to two times the sum of host atomic radius $2R$ plus twice the radius of the interstitial atom $2r$; i.e.,

$$\text{Unit cell edge length} = 2R + 2r$$

Now, equating these two unit cell edge length expressions, we get

$$2R + 2r = \frac{4R}{\sqrt{3}}$$

and solving for $r$ in terms of $R$

$$2r = \frac{4R}{\sqrt{3}} - 2R = \left(\frac{2}{\sqrt{3}} - 1\right)(2R)$$

or

$$r = \left(\frac{2}{\sqrt{3}} - 1\right)R = 0.155R$$

**Concept Check 4.1** Is it possible for three or more elements to form a solid solution? Explain your answer.

[*The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister (Student Companion Site).*]

**Concept Check 4.2** Explain why complete solid solubility may occur for substitutional solid solutions but not for interstitial solid solutions.

[*The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister (Student Companion Site).*]

## 4.4 SPECIFICATION OF COMPOSITION

**composition**

**weight percent**

It is often necessary to express the **composition** (or *concentration*)[5] of an alloy in terms of its constituent elements. The two most common ways to specify composition are weight (or mass) percent and atom percent. The basis for **weight percent** (wt%) is the weight of a particular element relative to the total alloy weight. For an alloy that

[5]The terms *composition* and *concentration* will be assumed to have the same meaning in this book (i.e., the relative content of a specific element or constituent in an alloy) and will be used interchangeably.

contains two hypothetical atoms denoted by 1 and 2, the concentration of 1 in wt%, $C_1$, is defined as

Computation of weight percent (for a two-element alloy)

$$C_1 = \frac{m_1}{m_1 + m_2} \times 100 \tag{4.3a}$$

where $m_1$ and $m_2$ represent the weight (or mass) of elements 1 and 2, respectively. The concentration of 2 is computed in an analogous manner.[6]

**atom percent**

The basis for **atom percent** (at%) calculations is the number of moles of an element in relation to the total moles of the elements in the alloy. The number of moles in some specified mass of a hypothetical element 1, $n_{m1}$, may be computed as follows:

$$n_{m1} = \frac{m_1'}{A_1} \tag{4.4}$$

Here, $m_1'$ and $A_1$ denote the mass (in grams) and atomic weight, respectively, for element 1.

Concentration in terms of atom percent of element 1 in an alloy containing element 1 and element 2 atoms, $C_1'$ is defined by[7]

Computation of atom percent (for a two-element alloy)

$$C_1' = \frac{n_{m1}}{n_{m1} + n_{m2}} \times 100 \tag{4.5a}$$

In like manner, the atom percent of element 2 is determined.[8]

Atom percent computations also can be carried out on the basis of the number of atoms instead of moles, because one mole of all substances contains the same number of atoms.

## Composition Conversions

Sometimes it is necessary to convert from one composition scheme to another—for example, from weight percent to atom percent. We next present equations for making these conversions in terms of the two hypothetical elements 1 and 2. Using the convention of the previous section (i.e., weight percents denoted by $C_1$ and $C_2$, atom percents by $C_1'$ and $C_2'$, and atomic weights as $A_1$ and $A_2$), we express these conversion equations as follows:

Conversion of weight percent to atom percent (for a two-element alloy)

$$C_1' = \frac{C_1 A_2}{C_1 A_2 + C_2 A_1} \times 100 \tag{4.6a}$$

$$C_2' = \frac{C_2 A_1}{C_1 A_2 + C_2 A_1} \times 100 \tag{4.6b}$$

---

[6]When an alloy contains more than two (say $n$) elements, Equation (4.3a) takes the form

$$C_1 = \frac{m_1}{m_1 + m_2 + m_3 + \cdots + m_n} \times 100 \tag{4.3b}$$

[7]In order to avoid confusion in notations and symbols being used in this section, we should point out that the prime (as in $C_1'$ and $m_1'$) is used to designate both composition in atom percent and mass of material in grams.

[8]When an alloy contains more than two (say $n$) elements, Equation (4.5a) takes the form

$$C_1' = \frac{n_{m1}}{n_{m1} + n_{m2} + n_{m3} + \cdots + n_{mn}} \times 100 \tag{4.5b}$$

Conversion of atom percent to weight percent (for a two-element alloy)

$$C_1 = \frac{C_1' A_1}{C_1' A_1 + C_2' A_2} \times 100 \tag{4.7a}$$

$$C_2 = \frac{C_2' A_2}{C_1' A_1 + C_2' A_2} \times 100 \tag{4.7b}$$

Because we are considering only two elements, computations involving the preceding equations are simplified when it is realized that

$$C_1 + C_2 = 100 \tag{4.8a}$$

$$C_1' + C_2' = 100 \tag{4.8b}$$

In addition, it sometimes becomes necessary to convert concentration from weight percent to mass of one component per unit volume of material (i.e., from units of wt% to kg/m$^3$); this latter composition scheme is often used in diffusion computations (Section 5.3). Concentrations in terms of this basis are denoted using a double prime (i.e., $C_1''$ and $C_2''$), and the relevant equations are as follows:

Conversion of weight percent to mass per unit volume (for a two-element alloy)

$$C_1'' = \left( \frac{C_1}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}} \right) \times 10^3 \tag{4.9a}$$

$$C_2'' = \left( \frac{C_2}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}} \right) \times 10^3 \tag{4.9b}$$

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For density $\rho$ in units of g/cm$^3$, these expressions yield $C_1''$ and $C_2''$ in kg/m$^3$.

Furthermore, on occasion we desire to determine the density and atomic weight of a binary alloy, given the composition in terms of either weight percent or atom percent. If we represent alloy density and atomic weight by $\rho_{ave}$ and $A_{ave}$, respectively, then

Computation of density (for a two-element metal alloy)

$$\rho_{ave} = \frac{100}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}} \tag{4.10a}$$

$$\rho_{ave} = \frac{C_1' A_1 + C_2' A_2}{\frac{C_1' A_1}{\rho_1} + \frac{C_2' A_2}{\rho_2}} \tag{4.10b}$$

Computation of atomic weight (for a two-element metal alloy)

$$A_{ave} = \frac{100}{\frac{C_1}{A_1} + \frac{C_2}{A_2}} \tag{4.11a}$$

$$A_{ave} = \frac{C_1' A_1 + C_2' A_2}{100} \tag{4.11b}$$

It should be noted that Equations 4.9 and 4.11 are not always exact. In their derivations, it is assumed that total alloy volume is exactly equal to the sum of the volumes of the individual elements. This normally is not the case for most alloys; however, it is a

reasonably valid assumption and does not lead to significant errors for dilute solutions and over composition ranges where solid solutions exist.

## EXAMPLE PROBLEM 4.3

### Derivation of Composition-Conversion Equation

Derive Equation 4.6a.

#### Solution

To simplify this derivation, we assume that masses are expressed in units of grams and denoted with a prime (e.g., $m_1'$). Furthermore, the total alloy mass (in grams) $M'$ is

$$M' = m_1' + m_2' \tag{4.12}$$

Using the definition of $C_1'$ (Equation 4.5a) and incorporating the expression for $n_{m1}$, Equation 4.4, and the analogous expression for $n_{m2}$ yields

$$C_1' = \frac{n_{m1}}{n_{m1} + n_{m2}} \times 100$$

$$= \frac{\dfrac{m_1'}{A_1}}{\dfrac{m_1'}{A_1} + \dfrac{m_2'}{A_2}} \times 100 \tag{4.13}$$

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Rearrangement of the mass-in-grams equivalent of Equation 4.3a leads to

$$m_1' = \frac{C_1 M'}{100} \tag{4.14}$$

Substitution of this expression and its $m_2'$ equivalent into Equation 4.13 gives

$$C_1' = \frac{\dfrac{C_1 M'}{100A_1}}{\dfrac{C_1 M'}{100A_1} + \dfrac{C_2 M'}{100A_2}} \times 100 \tag{4.15}$$

Upon simplification, we have

$$C_1' = \frac{C_1 A_2}{C_1 A_2 + C_2 A_1} \times 100$$

which is identical to Equation 4.6a.

## EXAMPLE PROBLEM 4.4

### Composition Conversion—From Weight Percent to Atom Percent

Determine the composition, in atom percent, of an alloy that consists of 97 wt% aluminum and 3 wt% copper.

#### Solution

If we denote the respective weight percent compositions as $C_{Al} = 97$ and $C_{Cu} = 3$, substitution into Equations 4.6a and 4.6b yields

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**Tutorial Video:** How to Convert from Atom Percent to Weight Percent

$$C'_{Al} = \frac{C_{Al}A_{Cu}}{C_{Al}A_{Cu} + C_{Cu}A_{Al}} \times 100$$
$$= \frac{(97)(63.55\ \text{g/mol})}{(97)(63.55\ \text{g/mol}) + (3)(26.98\ \text{g/mol})} \times 100$$
$$= 98.7\ \text{at\%}$$

and

$$C'_{Cu} = \frac{C_{Cu}A_{Al}}{C_{Cu}A_{Al} + C_{Al}A_{Cu}} \times 100$$
$$= \frac{(3)(26.98\ \text{g/mol})}{(3)(26.98\ \text{g/mol}) + (97)(63.55\ \text{g/mol})} \times 100$$
$$= 1.30\ \text{at\%}$$

# Miscellaneous Imperfections

## 4.5 DISLOCATIONS—LINEAR DEFECTS

**edge dislocation**
**dislocation line**



**WileyPLUS: VMSE** Edge

A *dislocation* is a linear or one-dimensional defect around which some of the atoms are misaligned. One type of dislocation is represented in Figure 4.4: an extra portion of a plane of atoms, or half-plane, the edge of which terminates within the crystal. This is termed an **edge dislocation;** it is a linear defect that centers on the line that is defined along the end of the extra half-plane of atoms. This is sometimes termed the **dislocation line,** which, for the edge dislocation in Figure 4.4, is perpendicular to the plane of the page. Within the region around the dislocation line there is some localized lattice distortion. The atoms above the dislocation line in Figure 4.4 are squeezed together, and those below are pulled apart; this is reflected in the slight curvature for the vertical planes of atoms as they bend around this extra half-plane. The magnitude of this distortion decreases with distance away from the dislocation line; at positions far removed, the crystal lattice is virtually perfect. Sometimes the edge dislocation in Figure 4.4 is represented by the symbol ⊥, which also indicates the position of the dislocation line. An edge dislocation may also be formed by an extra half-plane of atoms that is included in the bottom portion of the crystal; its designation is a ⊤.

**screw dislocation**

Another type of dislocation, called a **screw dislocation,** may be thought of as being formed by a shear stress that is applied to produce the distortion shown in Figure 4.5*a*: The

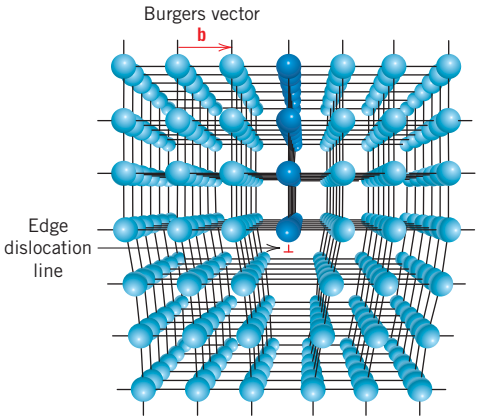
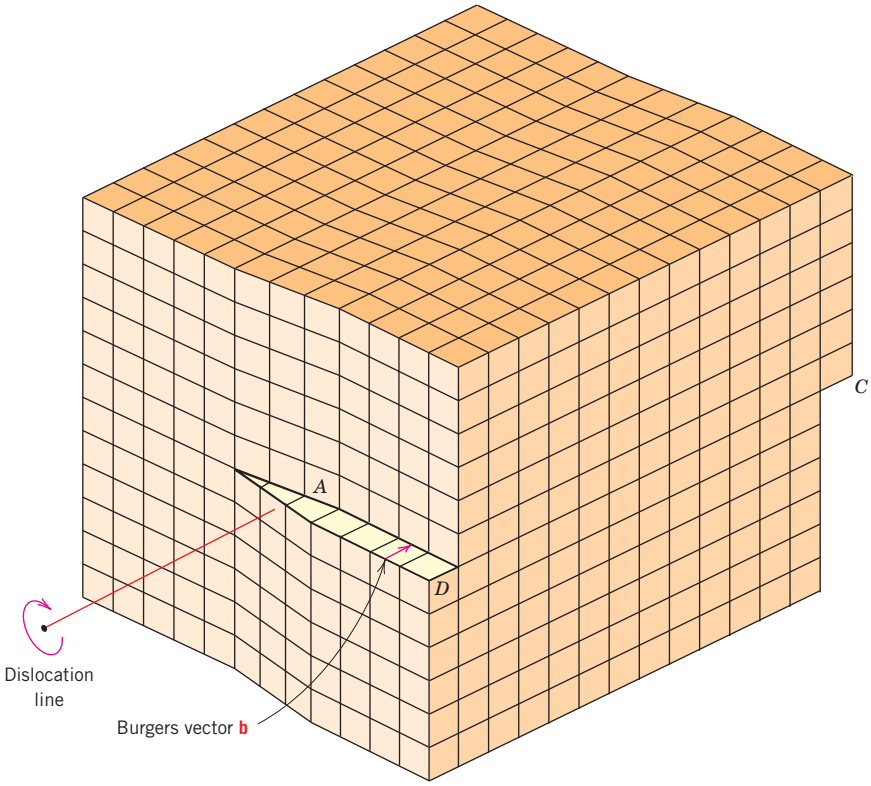



**Figure 4.4** The atom positions around an edge dislocation; extra half-plane of atoms shown in perspective.

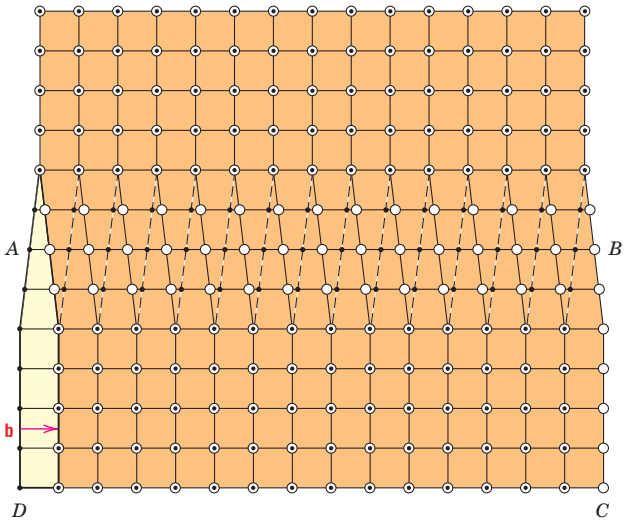


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Screw

upper front region of the crystal is shifted one atomic distance to the right relative to the bottom portion. The atomic distortion associated with a screw dislocation is also linear and along a dislocation line, line $AB$ in Figure 4.5*b*. The screw dislocation derives its name from the spiral or helical path or ramp that is traced around the dislocation line by the atomic planes of atoms. Sometimes the symbol ↻ is used to designate a screw dislocation.




(a)




(b)

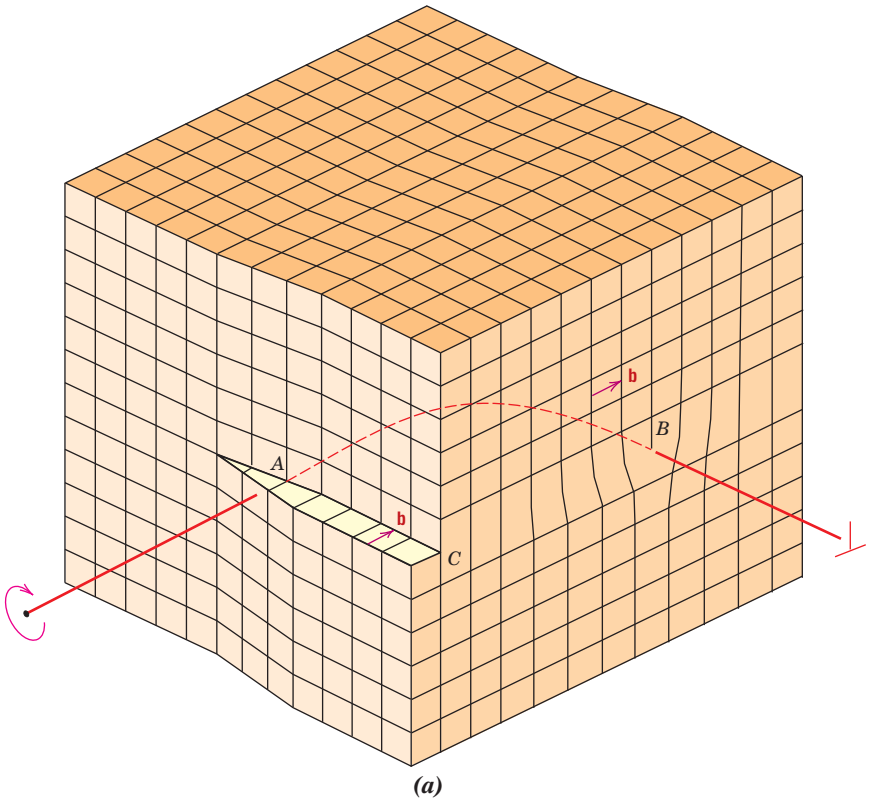
**Figure 4.5** (*a*) A screw dislocation within a crystal. (*b*) The screw dislocation in (*a*) as viewed from above. The dislocation line extends along line $AB$. Atom positions above the slip plane are designated by open circles, those below by solid circles.
[Figure (*b*) from W. T. Read, Jr., *Dislocations in Crystals,* McGraw-Hill Book Company, New York, NY, 1953.]

**mixed dislocation**



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Mixed

Most dislocations found in crystalline materials are probably neither pure edge nor pure screw but exhibit components of both types; these are termed **mixed dislocations.** All three dislocation types are represented schematically in Figure 4.6; the lattice distortion that is produced away from the two faces is mixed, having varying degrees of screw and edge character.



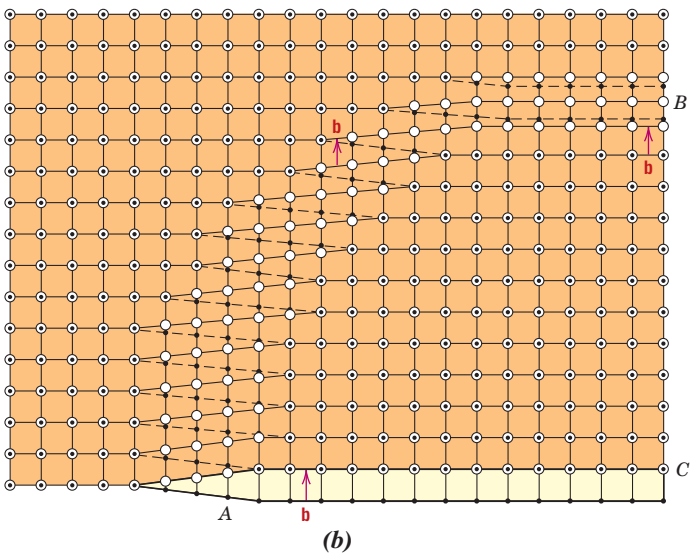




**Figure 4.6** (*a*) Schematic representation of a dislocation that has edge, screw, and mixed character. (*b*) Top view, where open circles denote atom positions above the slip plane, and solid circles, atom positions below. At point *A*, the dislocation is pure screw, while at point *B*, it is pure edge. For regions in between where there is curvature in the dislocation line, the character is mixed edge and screw.
[Figure (*b*) from W. T. Read, Jr., *Dislocations in Crystals,* McGraw-Hill Book Company, New York, NY, 1953.]